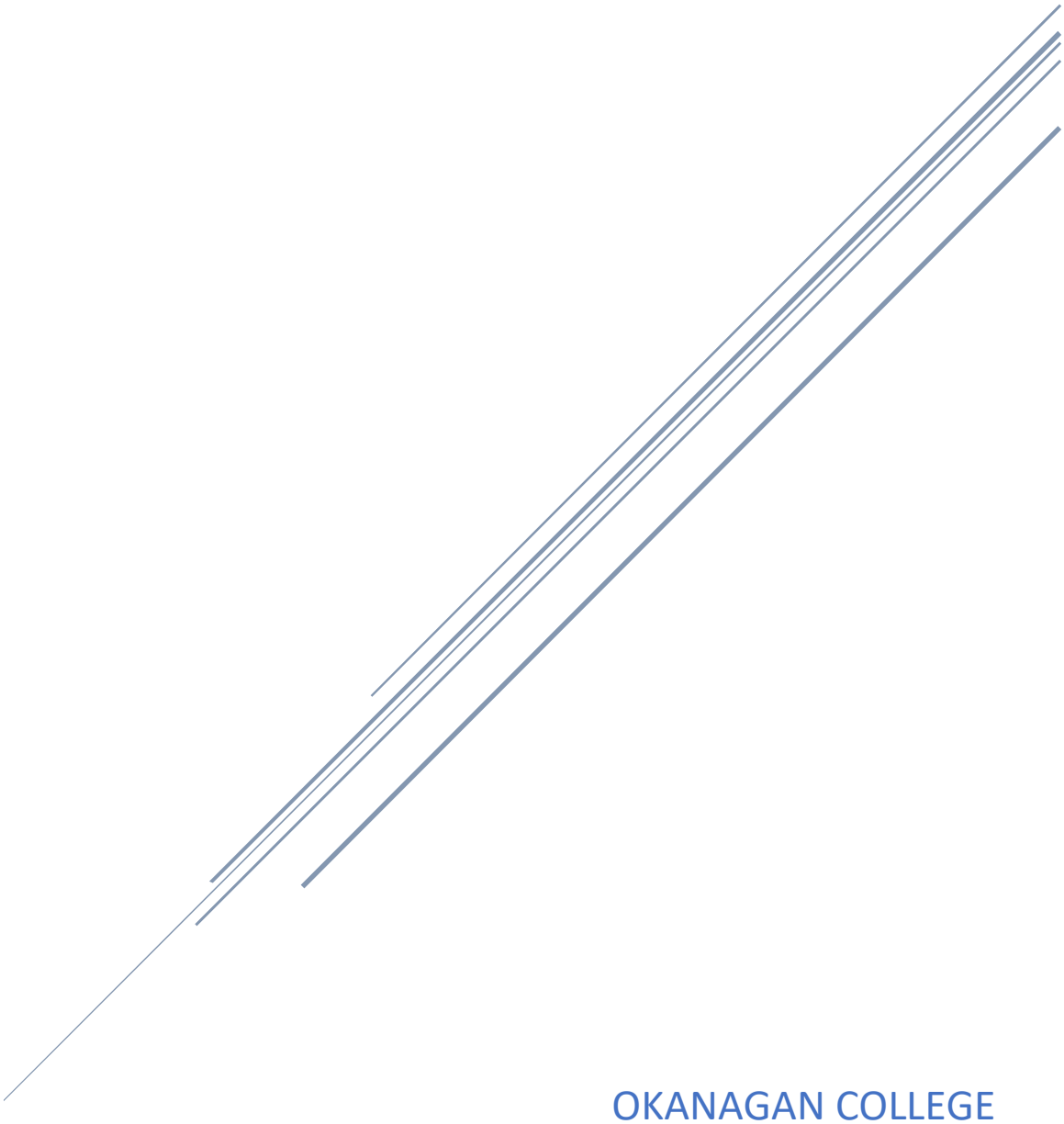
PREDICTING HOSPITAL LENGTH OF STAY USING GAMMA REGRESSION

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OKANAGAN COLLEGE

STAT 311

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1. Introduction

This report models the length of stay for patients admitted to hospital using a Gamma regression model. The objective is to estimate how long a patient is expected to remain in hospital based on 7 predictors: their age, the number of chronic conditions they have, whether the admission was elective or an emergency, their sex, and whether they were admitted on a weekend, severity and insurance status. The analysis is based on a sample of 420 patients. Length of stay ranges from 0.2 to 22.1 days, with a median LOS of 3.6 days and a mean of 4.61 days.

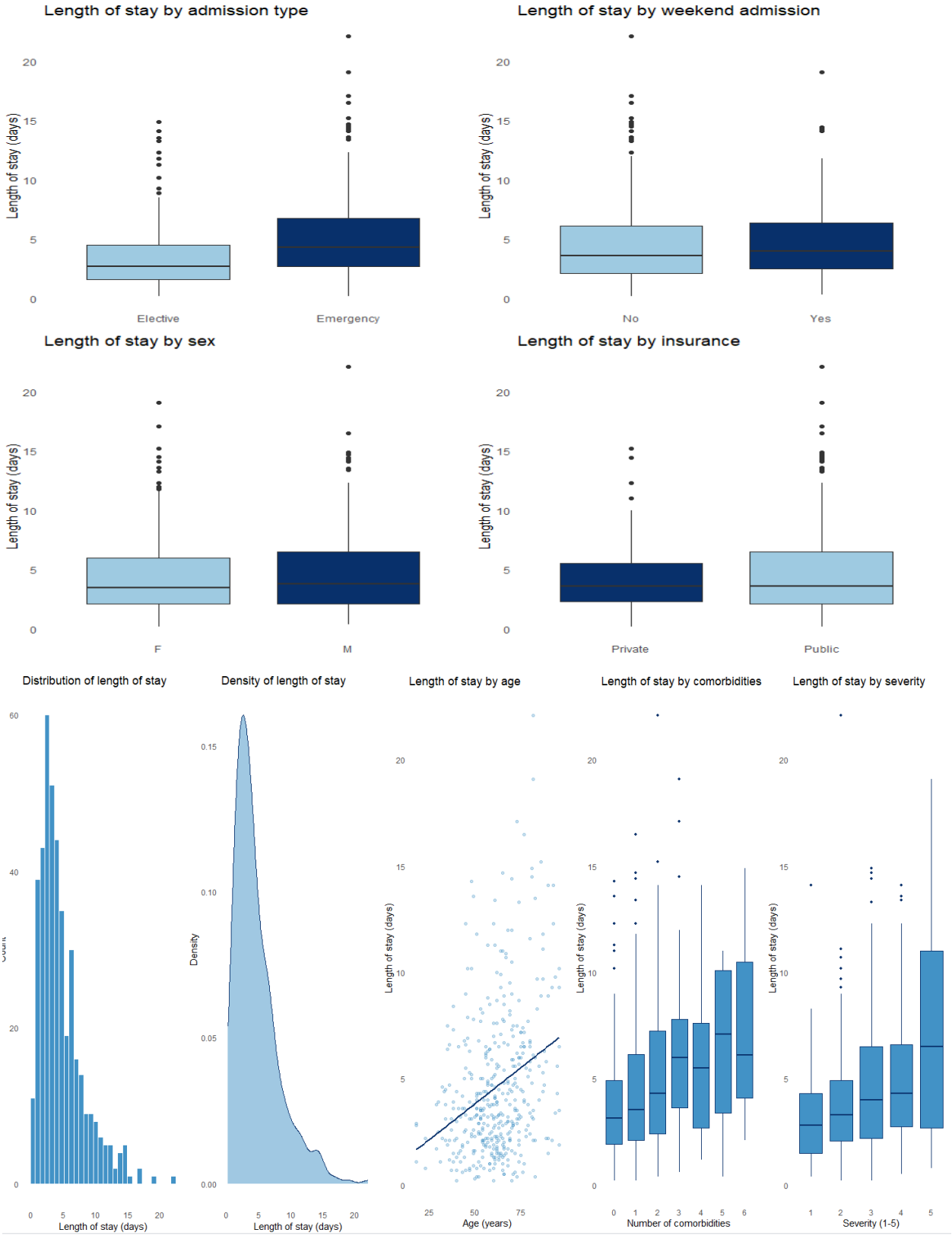
Ordinary Least Squares cannot model length of stay appropriately because length of stay cannot be negative, and its distribution is heavily right-skewed; most patients are discharged within a few days, while a small number remain for several weeks. Under these conditions OLS violates the assumption of constant variance. It can also produce negative predictions, which are impossible for a length of stay. Working with the log of length of stay does not resolve this either, because lm(log(y) ~ x) models the mean of the logged response, and exponentiating that result returns something closer to the median than the mean.

A Gamma model addresses all three problems. It assumes the response is strictly positive and continuous, it explicitly allows the variance to grow with the mean through the relationship $\text{Var}(Y) = \varphi\mu^2$, and because the model is fitted within the GLM framework, back-transforming returns the mean on the original scale. This makes Gamma the natural choice for right-skewed, strictly positive continuous data.

1.1 Model Assumptions

- Continuous, strictly positive response
- The conditional variance grows with the square of the mean.
- Independent observations. Each patient contributes one observation, with no clustering or repeated measures.
- No severe multicollinearity among predictors.

2. Exploratory Data Analysis



- Length of stay's mean exceeds the median and the long right tail in the histogram confirm a right-skewed, a strictly positive response which is a condition that Gamma regression requires.
- LOS rises steadily with age, from roughly 2 days near 20 years to 6–7 days near 90years
- Comorbidities Median LOS climbs from about 3 days at zero conditions to 6 days at five or six, with the spread widening at the higher counts.
- Severity is the steepest gradient in the data with a median LOS that rises monotonically from about 3 days at severity 1 to 6.5 days at severity 5, and the variance widens as the mean grows, the variance-mean pattern that the Gamma family assumes.
- Emergency admissions stay longer than the elective admission
- Weekend admissions have a modestly higher median LOS (≈4.3 vs ≈3.4 days).
- Sex. Near-balanced,214 female, 206 males. Medians are close (F ≈3.4, M ≈3.8), so little separation expected.
- Insurance. 304 public, 116 private. Public patients show a lower median LOS (≈3.5) than private (≈4.3), which runs opposite to expectation and may reflect confounding with severity or comorbidity to be revisited after fitting.

3. Correlation

	los_days	age	severity	comorbid.
los_days	1	0.297	0.279	0.251
age	0.297	1	0.034	0.408
severity	0.279	0.034	1	-0.007
comorbidities	0.251	0.408	-0.007	1

For collinearity, age and comorbidities correlate at 0.408; no other numeric pair exceeds 0.30. Severity is effectively independent of both (0.034 with age, −0.007 with comorbidities), so its coefficient should be estimated cleanly.

4. Model Building

4.1 Full Model

Coefficient	Estimate	Std. Error	t value	Pr(>|t|)	x= exp(Estimate)	Change(x-1)	Signif.
(Intercept)	0.005421	0.165533	0.033	0.97389			
age	0.009822	0.002266	4.335	1.83E-05	1.0099	0.99%	***
comorbidities	0.086032	0.02941	2.925	0.00363	1.0898	8.98%	**
admissionEmergency	0.323428	0.065574	4.932	1.18E-06	1.3819	38.19%	***
sexM	0.061575	0.063094	0.976	0.32967	1.0635	6.35%	
weekendYes	0.056955	0.069428	0.82	0.4125	1.0586	5.86%	
severity	0.160774	0.026261	6.122	2.15E-09	1.1744	17.44%	
insurancePublic	0.078726	0.070792	1.112	0.26676	1.0819	8.19%	***

The model;

$\log(\hat{\mu}) = 0.0054+0.0098age+0.0860comorbidities+0.3234emergency+0.1608severity$

Four predictors are significant. Severity is the strongest (t = 6.12): each additional level multiplies expected stay by 1.17, holding all else constant. Emergency admission multiplies it by 1.38. Age adds about 1% per year, meaning ten years of age raises expected stay roughly 10%. Each additional comorbidity multiplies expected stay by 1.09, so three conditions against none is about 29% longer (e^3×0.086032=e^0.258096=1.2945).

Three predictors are not significant. Sex (p = 0.33), weekend (p = 0.41) and insurance (p = 0.27) show no significant effect given the others. Notably, insurance reverses sign relative to the EDA public patients appeared to have *shorter* stays before adjustment but *longer* after, indicating the raw difference reflected could have confounding effect by age and morbidity rather than insurance itself.

The null deviance of 242.48 falls to 190.46, a reduction of 52.0 across the seven predictors.

4.2 Reduced Model

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	0.105904	0.151804	0.698	0.4858
age	0.009741	0.002257	4.317	1.98e-05 ***
comorbidities	0.087316	0.029345	2.976	0.0031 **
admissionEmergency	0.326531	0.065692	4.971	9.78e-07 ***
severity	0.162933	0.026286	6.198	1.38e-09 ***

The model;

$\log(\hat{\mu}) = 0.1059 + 0.0097\text{age} + 0.0873\text{comorbidities} + 0.3265\text{emergency} + 0.1629\text{severity}$

The reduced model retained all the predictors; age, comorbidities, admission type and severity. AIC improves from 1950.9 to 1947.8 despite three fewer parameters, and the residual deviance rises by only 1.25 across 415 degrees of freedom which is negligible. This indicates the three omitted terms contributed almost no explanatory power. The coefficients of the retained predictors shifted by less than 2% after removal, age from 0.01257 to 0.00974 being the largest change, reflecting that comorbidities and age share some explanatory power (they correlate at 0.408). No retained coefficient changed direction or lost significance, which confirms none of the removed variables was acting as a confounder for the others.

5. Models Comparison

Model	AIC	Key assumption	Result
Normal LM	2149.08	Constant variance	Violated - residuals fan out
Poisson	2086.76	Variance = mean	Violated - overdispersed ($\chi^2/df = 1.89$)
Gamma	1947.83	Variance = $\varphi\mu^2$	Consistent with the data

The Gamma model's AIC is 139 points lower than the Poisson and 201 points lower than the normal linear model. The normal model assumes a single constant variance across all fitted values, which is untenable when predicted stays range from 0.6 to 9.6 days; the Poisson assumes the variance equals the mean, and the data are over dispersed by a factor of 1.89. The Gamma family allows the variance to grow with the mean, which is the pattern the residuals display.

All three models identify the same four predictors as significant, with the same signs and ranking. The difference is in how well each describes the distribution of length of stay, and on that the Gamma model is clearly preferred.

6. Post Hoc Analysis

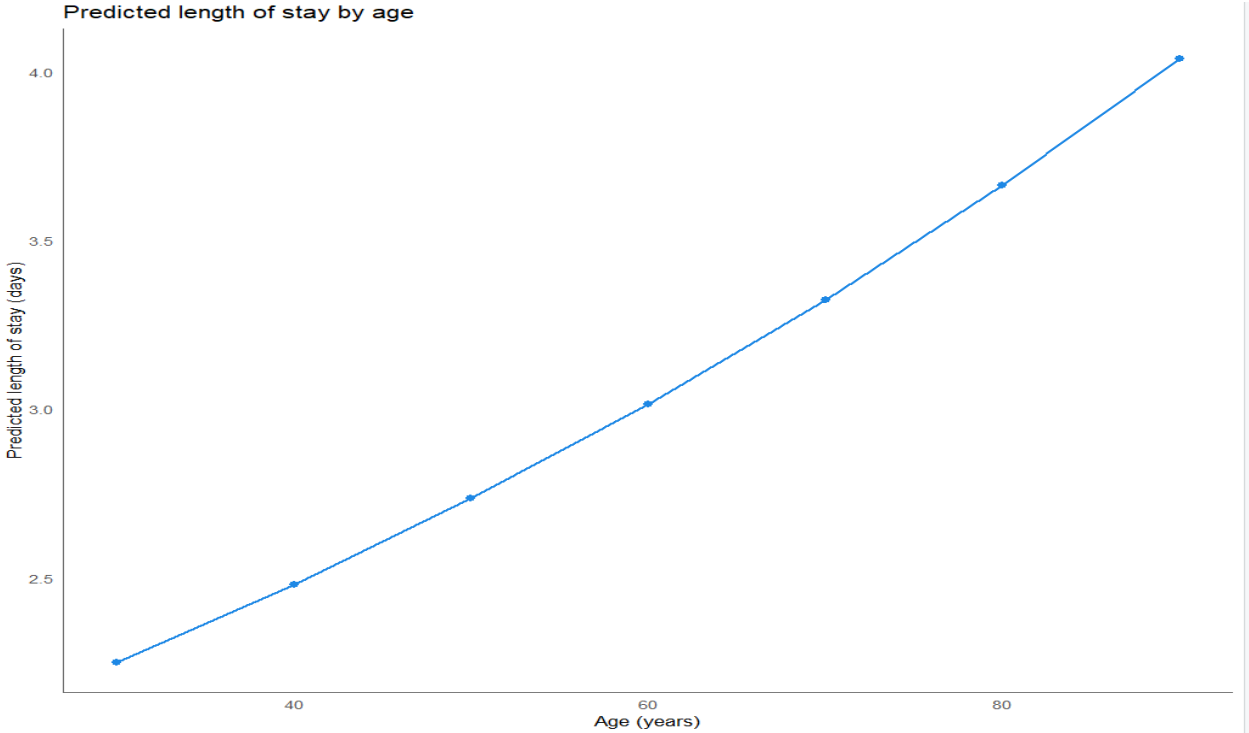
Reference patient: age 60, one comorbidity, elective admission, severity 2.

Profile	Age	Comorbidities	Admission	Severity	Predicted LOS (days)
Reference	60	1	Elective	2	3.015
Ten years older	70	1	Elective	2	3.323
(+1)Severity	60	1	Elective	3	3.548
Emergency	60	1	Emergency	2	4.179
Emergency and higher severity	60	1	Emergency	4	5.789

The table reports expected length of stay for a patient aged 60 with one comorbidity, varying one or two characteristics at a time. The reference patient, elective admission at severity 2 is predicted to stay 3.02 days, which sits slightly below the sample median of 3.6 days, consistent with a comparatively mild case. Each change moves the prediction by a fixed multiplier rather than a fixed number of days, as the log link requires, ten additional years of age raises the expected stay to 3.32 days (+10.2%), one severity level to 3.55 days (+17.7%), and emergency admission to 4.18 days (+38.6%). The final row combines emergency admission with severity 4 and predicts 5.79 days which is an increase of 92% over the reference. This is nearly double rather than the sum of the individual effects (10% + 18% + 39%), because effects compound multiplicatively: $1.386 \times 1.174^2 = 1.91$. The table therefore illustrates both the direction and the compounding behaviour of the fitted model.

6.1 Across age (one comorbidity, elective, severity 2)

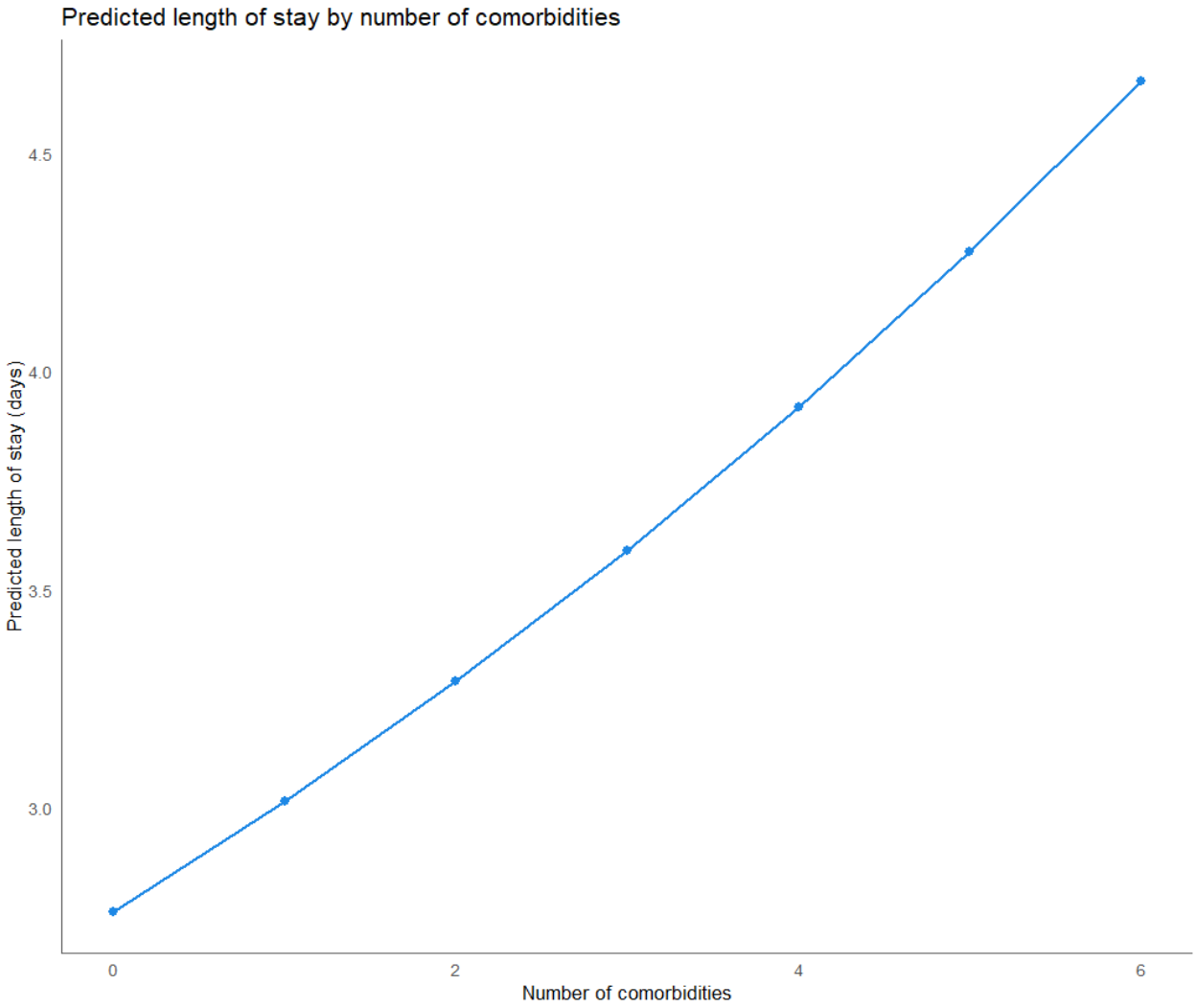
Age	30	40	50	60	70	80	90
Predicted LOS (days)	2.251	2.481	2.735	3.015	3.015	3.663	4.038



Predicted length of stay rises from 2.25 days at age 30 to 4.04 days at age 90 , a 79% increase holding comorbidities at 1, severity at 2 and elective admission. The curve climbs steadily because the model applies a constant multiplier of 1.0098 per year: a fixed 1% of an increasing baseline produces increasingly larger day-increments. This is consistent with the EDA, where age showed the strongest bivariate correlation with length of stay among the numeric predictors (0.297). Since age and comorbidity correlate at 0.408, the age coefficient is the effect of age *given* comorbidity count rather than age alone and older patients carry more conditions, so the combined effect of both is larger than either coefficient suggests.

6.2 Across comorbidities (age 60, elective, severity 2)

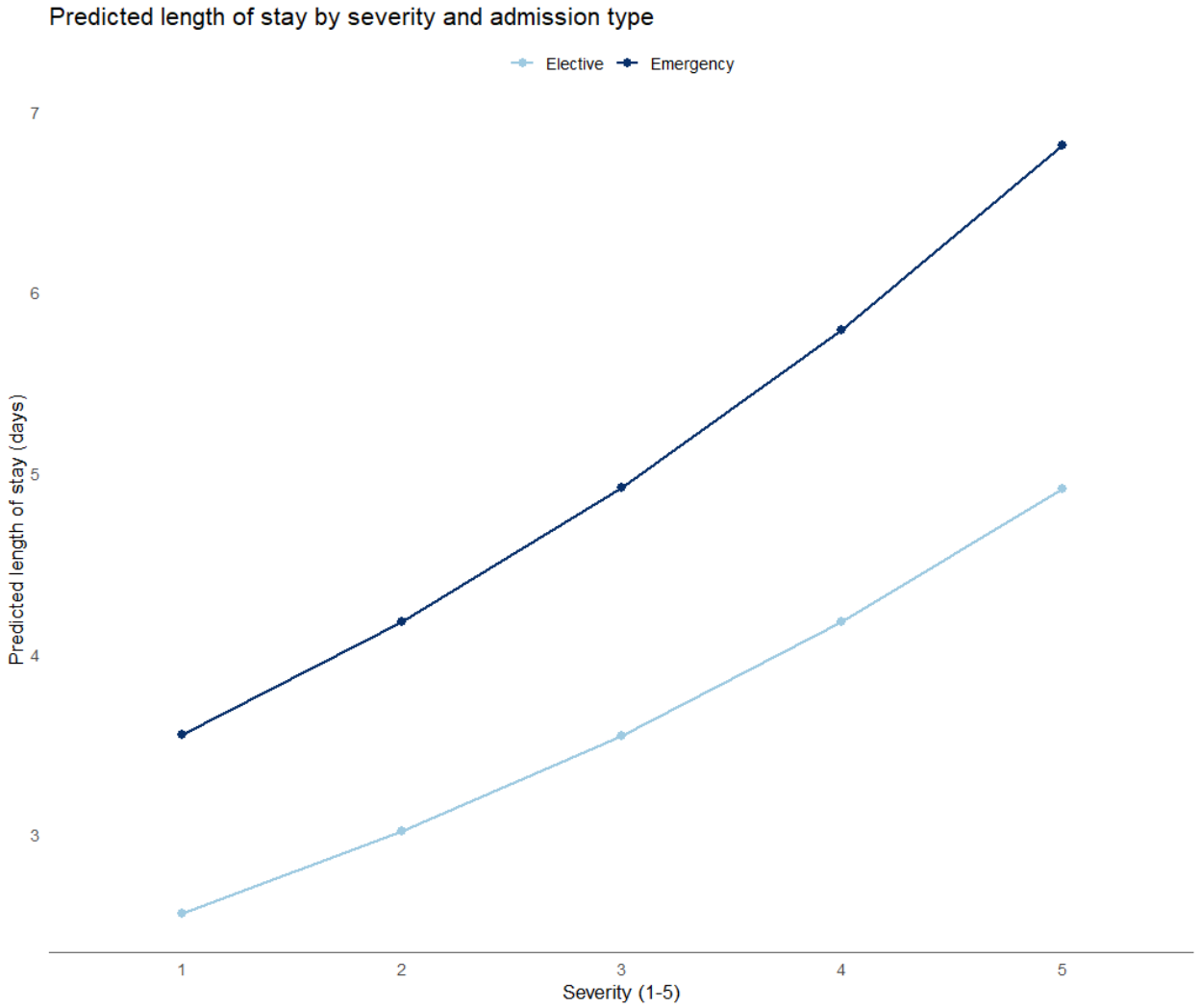
comorbidities	0	1	2	3	4	5	6
Predicted LOS (days)	2.763	3.015	3.29	3.59	3.918	4.275	4.665



Each additional chronic condition multiplies expected length of stay by 1.091, about 9.1%, holding age, admission type and severity fixed. The effect is statistically robust ($p = 0.003$), and it is the third-largest in the model, behind severity and emergency admission. This is consistent with the exploratory analysis, where median length of stay climbed from about 3 days at zero conditions to 6 days at five or six, with the spread widening at higher counts.

6.3 Across severity (age 60, one comorbidity, elective)

Severity	1	2	3	4	5
Predicted LOS (days)	2.562	3.015	3.548	4.176	4.915



The figure plots predicted length of stay against severity separately for elective and emergency admissions, holding age at 60 and comorbidities at one. Both curves rise, but they rise at an increasing rate in days — because the fitted model is multiplicative, each additional severity level multiplies expected stay by a constant 1.177 rather than adding a fixed number of days. The gap between the curves also widens as severity increases, from roughly 1.1 days at severity 1 to 1.9 days at severity 5, since the emergency multiplier of 1.386 operates on an increasingly larger baseline. A model with an identity link would produce parallel lines instead. The graph therefore illustrates the two strongest predictors acting together: an elective patient at severity 1 is expected to stay 2.56 days, against approximately 6.8 days for an emergency patient at severity 5.

7. Findings

- Severity is the strongest predictor of length of stay. Each additional level multiplies expected stay by 1.177, a 17.7% increase and the model's largest t value (6.12) belongs to it. Median stay rose from about 3 days at severity 1 to 6.5 days at severity 5 in the EDA, and the coefficient confirms it.
- Emergency admissions stay roughly 39% longer than elective. The multiplier is 1.386, making admission type the second-strongest effect. In the worked example the reference patient's expected stay rises from 3.02 to 4.18 days from admission type alone.
  Age's effect is small per year but substantial over a lifetime. At 1% per year, it looks negligible, yet ten years multiplies expected stay by 1.10 and the range from 30 to 90 raises it from 2.25 to 4.04 days, a 79% increase.
- Effects compound rather than add, and administrative variables do not predict stay. A patient who is both an emergency admission and two severity levels above reference is expected to stay 1.91 times as long, not the sum of individual effects. Meanwhile sex ($p = 0.379$), weekend admission ($p = 0.474$) and insurance ($p = 0.184$) show no significant effect once clinical factors are controlled, and insurance reverses direction, revealing that its raw association reflected case mix rather than insurance itself.

8. Recommendations

- Use severity and admission type to flag expected long-stay patients at the point of admission. These are the two strongest predictors, and both are known immediately, so they can drive early discharge planning and resource allocation without waiting for further information.
- Plan for rising average length of stay as the population ages. Age adds roughly 10% per decade and travels with comorbidity, so demand will grow even if severity and admission mix remain constant, a demographic effect that capacity planning should anticipate.
- Extend the data before applying the model operationally. Surgical status was absent from the dataset and two of the specified inputs proved non-significant, so collecting surgical and specialty information would likely sharpen the model, since clinical variables clearly dominate administrative ones here.

9. Appendix

9.1 Summary Statistic

9.1.1 Admission

A tibble: 2 × 4

admission	n	mean_los	median_los
<fct>	<int>	<dbl>	<dbl>
1 Elective	153	3.58	2.7
2 Emergency	267	5.21	4.3

9.1.2 Weekend

A tibble: 2 × 4

weekend	n	mean_los	median_los
<fct>	<int>	<dbl>	<dbl>
1 No	300	4.55	3.6
2 Yes	120	4.76	4

9.1.3 Commorbidities

comorbidities	n	mean_los
<int>	<int>	<dbl>
1 0	154	3.78
2 1	138	4.45
3 2	83	5.3
4 3	27	6.85
5 4	11	6.11
6 5	4	6.4
7 6	3	7.7

9.1.4 Severity

severity	n	mean_los
<int>	*<int>*	*<dbl>*
1	1	61	3.18
2	2	124	4.01
3	3	111	4.85
4	4	83	5.03
5	5	41	7.07

9.1.5 Sex

sex	n	mean_los
<fct>	*<int>*	*<dbl>*
1 F	214	4.43
2 M	206	4.8

9.1.6 Insurance

insurance	n	mean_los
<fct>	*<int>*	*<dbl>*
1 Private	116	4.24
2 Public	304	4.75

9.2 Full Model output (Gamma)

9.2.1 summary(m_full)

Call:

glm(formula = los_days ~ age + comorbidities + admission + sex +

weekend + severity + insurance, family = Gamma(link = "log"),

data = Lab3)

Coefficients:

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	0.005421	0.165533	0.033	0.97389
age	0.009822	0.002266	4.335	1.83e-05 ***
comorbidities	0.086032	0.029410	2.925	0.00363 **
admissionEmergency	0.323428	0.065574	4.932	1.18e-06 ***
sexM	0.061575	0.063094	0.976	0.32967
weekendYes	0.056955	0.069428	0.820	0.41250
severity	0.160774	0.026261	6.122	2.15e-09 ***
insurancePublic	0.078726	0.070792	1.112	0.26676

---Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Gamma family taken to be 0.4110459)

Null deviance: 242.48 on 419 degrees of freedom

Residual deviance: 190.46 on 412 degrees of freedom

AIC: 1950.9

Number of Fisher Scoring iterations: 5

9.2.2 Reduced Model output (Gamma)

Coefficients:

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	0.105904	0.151804	0.698	0.4858
age	0.009741	0.002257	4.317	1.98e-05 ***
comorbidities	0.087316	0.029345	2.976	0.0031 **
admissionEmergency	0.326531	0.065692	4.971	9.78e-07 ***
severity	0.162933	0.026286	6.198	1.38e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Gamma family taken to be 0.4125854)

Null deviance: 242.48 on 419 degrees of freedom

Residual deviance: 191.71 on 415 degrees of freedom and AIC: 1947.8

9.2.3 Poison Model

Coefficients:

Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	0.070299	0.118803	0.592	0.554
age	0.010779	0.001666	6.469	9.89e-11 ***
comorbidities	0.074761	0.019176	3.899	9.68e-05 ***
admissionEmergency	0.310890	0.050967	6.100	1.06e-09 ***
severity	0.161198	0.018817	8.567	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 995.57 on 419 degrees of freedom

Residual deviance: 766.58 on 415 degrees of freedom

AIC: 2086.8

9.2.4 Linear Regression Model

Residuals:

Min 1Q Median 3Q Max

-7.5044 -2.0556 -0.5463 1.3731 16.2840

Coefficients:

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	-1.77262	0.73248	-2.420	0.01595 *
age	0.04743	0.01089	4.356	1.67e-05 ***
comorbidities	0.44404	0.14159	3.136	0.00183 **
admissionEmergency	1.27322	0.31698	4.017	7.00e-05 ***
severity	0.76893	0.12684	6.062	3.02e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.099 on 415 degrees of freedom

Multiple R-squared: 0.2132, Adjusted R-squared: 0.2056

F-statistic: 28.11 on 4 and 415 DF, p-value: < 2.2e-16